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Inventor(s)

Vivanco; Daniel et al.

SYSTEMS AND METHODS FOR TRAFFIC MANAGEMENT BASED ON UPLINK POWER REQUIREMENTS IN NON- TERRESTRIAL NETWORKS

Abstract

Aspects of the subject disclosure may include, for example, obtaining data indicative of a location of a satellite; obtaining data indicative of locations of each of a plurality of mobile devices, battery states of each of the mobile devices, and transmit powers that are required for the mobile devices to upload traffic to a wireless terrestrial network; generating, based upon the location of the satellite and upon the locations of the mobile devices, data indicative of predicted transmit powers that would be required for each of the mobile devices to upload traffic to the satellite; selecting, based upon the battery states of each of the mobile devices, upon the transmit powers that are required for each of the mobile devices to upload traffic to the wireless terrestrial network, and upon the predicted transmit powers that would be required for the mobile devices to upload traffic to the satellite, at least one mobile device of the plurality of mobile devices to be instructed to upload traffic to the satellite. Other embodiments are disclosed.

Inventors: Vivanco; Daniel (Ashburn, VA), Beppler; David Ross (Duluth, GA)

Applicant: AT&T Intellectual Property I, L.P. (Atlanta, GA); AT&T Technical Services Company, Inc. (Vienna, VA)

Family ID: 96949670

Assignee: AT&T Intellectual Property I, L.P. (Atlanta, GA); AT&T Technical Services Company, Inc. (Vienna, VA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application is related to U.S. application Ser. No. 18/470,125, filed Sep. 19, 2023. All sections of the aforementioned application(s) and/or patent(s) are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The subject disclosure relates to systems and methods for traffic management based on uplink power requirements in non-terrestrial networks.

BACKGROUND

[0003] A cellular service provider (CSP) with a terrestrial cellular network (TN) may own a direct cellular-to-satellite non-terrestrial network in addition to the terrestrial network. A cellular-to-satellite non-terrestrial network (NTN) can create a direct connection from a customer's/subscriber's garden variety cellular telephone using LTE, 5G, GSM, UMTS, 6G (or other commercially-available cellular technology user equipment (UE)) to a satellite (wherein the satellite must use frequency bands that the UE is already designed to communicate with, and must use either unlicensed bands or bands that are licensed to the CSP).

[0004] Satellite cells may be used to provide additional coverage or capacity to terrestrial cells. Satellite cells may have the ability to use one or more cells, and to operate at different frequency bands (e.g., B5,B14). A CSP (or wireless operator) may have the ability to mandate satellite cells to change frequency bands (e.g. to avoid interference with terrestrial cells, which may operate in the same frequency band).

[0005] A CSP can use satellite cells to provide additional service capacity to specific areas at specific hours of the day (e.g., busy hour). Also, a CSP may have the ability to control the coverage of the satellite cells and/or schedule the actuation of the satellite cells (e.g., in such a way that satellite cells orbiting over the congested terrestrial RAN areas provide coverage at their corresponding busy times).

[0006] A CSP is typically aware of the time and duration that the satellite cells will cover the congested areas. The CSP is also typically aware of the frequency bands used in the terrestrial cells, and also is aware of the phone capabilities and bands that they support (a given CSP may use different frequency bands in different locations).

[0007] Accordingly, a CSP can use certain conventional traffic management techniques to offload traffic from a terrestrial cellular network to a non-terrestrial network (such conventional traffic management techniques are typically based on loading conditions).

[0008] With reference now to the satellites themselves, it is noted that a CSP may deploy several satellite cells in a batch (constellation), and each satellite cell may have a predefined coverage area (e.g., 50 Km radius). Satellite cells may be managed in a specific formation in an effort to provide continuous coverage to terrestrial UEs. A constellation of satellite cells can typically provide larger continuous coverage based on the number of satellite cells in the group.

[0009] In addition, a CSP may have several constellations of satellite cells and the constellations may have gaps in between. Therefore, there may be times at which a terrestrial UE will not detect any satellite.

[0010] With reference now to certain conventional terrestrial cellular network power management mechanisms, a maximum allowed UE transmit power is traditionally set in LTE/5G terminals by

“P_powerclass”. “P_powerclass” is the configured maximum UE output power, which varies based on UE Power Class. UE-Power.Class.1 allows for output power levels of +31 dBm, UE-Power.Class.2 allows for output power levels of 26 dBm, and UE-Power.Class.3 allows for power levels of 23 dbm.

[0011] Further, an LTE UE transmit power control mechanism traditionally adjusts UE transmit power (P.sub.PUSCH) based on traffic demand and UE location with respect to the serving cell. P.sub.PUSCH (Physical Uplink Shared Channel Power) can be set up to “P_powerclass”. A UE engaged into heavy uplink traffic, located far away from serving cell may get P.sub.PUSCH set to a high value, which may drain battery quickly (especially for small devices such as smartphones or wearable devices).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0013] FIG. 1 is a block diagram illustrating an example, non-limiting embodiment of a communication network in accordance with various aspects described herein.

[0014] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system (which can function fully or partially within the communication network of FIG. 1) in accordance with various aspects described herein.

[0015] FIG. 2B depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0016] FIG. 2C depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0017] FIG. 2D depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0018] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0019] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0020] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0021] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0022] The subject disclosure describes, among other things, illustrative embodiments for traffic management based on uplink power requirements in non-terrestrial networks. Other embodiments are described in the subject disclosure.

[0023] One or more aspects of the subject disclosure include a device, comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: obtaining first data associated with a satellite of a wireless non-terrestrial network, wherein the first data indicates a location of the satellite; obtaining second data associated with a plurality of mobile devices that are respectively attached to a wireless terrestrial network, wherein the second data indicates a respective location of each of the plurality of mobile devices, a respective battery state of each of the plurality of mobile devices, and a respective transmit power that is required for each of the plurality of mobile devices to upload traffic to the wireless terrestrial network; generating, based upon the location of the satellite indicated by the first data and upon the respective location

of each of the plurality of mobile devices indicated by the second data, third data that indicates respective predicted transmit power that would be required for each of the plurality of mobile devices to upload traffic to the satellite; selecting, based upon the respective battery state of each of the plurality of mobile devices, upon the respective transmit power that is required for each of the plurality of mobile devices to upload traffic to the wireless terrestrial network, and upon the respective predicted transmit power that would be required for each of the plurality of mobile devices to upload traffic to the satellite, at least one mobile device of the plurality of mobile devices to be instructed to upload traffic to the satellite, wherein the selecting results in at least one selected mobile device; and responsive to the selecting, sending a command to the at least one selected mobile device to upload the traffic to the satellite instead of to the wireless terrestrial network.

[0024] One or more aspects of the subject disclosure include a non-transitory machine-readable medium comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: obtaining first data associated with a first end-user device that is attached to a wireless terrestrial network, wherein the first data indicates a first location of the first end-user device and a first configuration of the first end-user device, and wherein the first location comprises a first current location, a first predicted future location, or any combination thereof; obtaining second data associated with a second end-user device that is attached to the wireless terrestrial network, wherein the second data indicates a second location of the second end-user device and a second configuration of the second end-user device, and wherein the second location comprises a second current location, a second predicted future location, or any combination thereof; obtaining third data associated with a satellite of a wireless non-terrestrial network, wherein the third data indicates a third location of the satellite, and wherein the third location comprises a third current location, a third predicted future location, or any combination thereof; generating, based upon the first location and upon the third location, fourth data that indicates a first predicted transmit power that would be required for the first end-user device to upload traffic to the satellite; generating, based upon the second location and upon the third location, fifth data that indicates a second predicted transmit power that would be required for the second end-user device to upload traffic to the satellite; and selecting either the first end-user device or the second end-user device to be commanded to communicate with the satellite, wherein the selecting results in a selected end-user device, and wherein the selecting is based upon the first configuration of the first end-user device, upon the second configuration of the second end-user device, upon the first predicted transmit power that would be required for the first end-user device to upload traffic to the satellite, and upon the second predicted transmit power that would be required for the second end-user device to upload traffic to the satellite.

[0025] One or more aspects of the subject disclosure include a method, comprising: obtaining, by a processing system comprising a processor, first information associated with a first end-user device that is attached to a wireless terrestrial network, wherein the first information indicates a first location of the first end-user device and a first battery state of the first end-user device, and wherein the first location comprises a first current location, a first predicted future location, or any combination thereof; obtaining, by the processing system, second information associated with a second end-user device that is attached to the wireless terrestrial network, wherein the second information indicates a second location of the second end-user device and a second battery state of the second end-user device, and wherein the second location comprises a second current location, a second predicted future location, or any combination thereof; obtaining, by the processing system, third information associated with a third end-user device that is attached to the wireless terrestrial network, wherein the third information indicates a third location of the second end-user device and a third battery state of the third end-user device, and wherein the third location comprises a third current location, a third predicted future location, or any combination thereof; obtaining, by the processing system, fourth information indicative of a fourth location of a low earth orbit (LEO)

satellite, wherein the LEO satellite is part of a constellation of LEO satellites forming a wireless non-terrestrial network, and wherein the fourth location comprises a fourth current location, a fourth predicted future location, or any combination thereof; generating, by the processing system, based upon the first location and upon the fourth location, fifth information indicative of a first predicted power that would be required for the first end-user device to carry out first uplink transmissions to the LEO satellite; generating, by the processing system, based upon the second location and upon the fourth location, sixth information indicative of a second predicted power that would be required for the second end-user device to carry out second uplink transmissions to the LEO satellite; generating, by the processing system, based upon the third location and upon the fourth location, seventh information indicative of a third predicted power that would be required for the third end-user device to carry out third uplink transmissions to the LEO satellite; and rank ordering, by the processing system, the first end-user device, the second end-user device and the third end-user device, wherein the rank ordering is based upon the first battery state, the first predicted power, the second battery state, the second predicted power, the third battery state, and the third predicted power, and wherein the rank ordering orders each of the first end-user device, the second end-user device, and the third end-user device with respect to being instructed to hand-over to the LEO satellite.

[0026] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part traffic management between a terrestrial network and a non-terrestrial network based upon user equipment battery state and uplink power requirements. In particular, a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

[0027] The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VoIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0028] In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0029] In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0030] In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

[0031] In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

[0032] In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0033] In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150, 152, 154, 156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0034] Referring now to FIG. **2A**, this shows a block diagram illustrating an example, non-limiting embodiment of a system **200** (which can function fully or partially within the communication network of FIG. **1**) in accordance with various aspects described herein. As seen in this figure, wireless terrestrial network **202** is configured for wireless communications with a plurality of devices **204A, 204B, 204C, 204D**. In various embodiments, the wireless terrestrial network **202** comprises a cellular network, a fifth generation (5G) cellular network, a sixth generation (6G) cellular network, a subsequent generation cellular network, one or more RANs, one or more eNBs, one or more gNBs, one or more SONs, one or more RICs, or any combination thereof. In various embodiments, each of devices **204A, 204B, 204C, 204D** comprises a respective wearable device, a respective smartwatch; a respective smartphone, a respective cellphone, a respective laptop computer, a respective notebook computer, a respective tablet, or any respective combination thereof.

[0035] Still referring to FIG. **2A**, it is seen that a plurality of satellites **206A, 206B, 206C, 206D, 206E** form a constellation (or batch) of satellite cells that can be used to provide larger/contiguous coverage to terrestrial UEs. In various embodiments, each of satellites **206A, 206B, 206C, 206D, 206E** can comprise a low earth orbit (LEO) satellite.

[0036] Still referring to FIG. **2A**, it is seen that server(s) **202A** can be part of wireless terrestrial network **202**. These server(s) **202A** can comprise hardware, firmware, and/or software for carrying out various traffic management techniques as described herein (for example, to control hand-over to one or more satellites). In various embodiments, the server(s) **202A** can be part of the wireless terrestrial network **202** and/or can be located separate (e.g., remotely) from the wireless terrestrial network **202**. Of course, while four devices (**204A-204D**) are shown in this figure, any desired number of such devices can be supported. Further, while five satellites (**206A-206E**) are shown in this figure, any desired number of satellites can be supported. In various embodiments, the server(s) **202A** can control hand-over of one or more devices from the wireless terrestrial network **202** to one of (or a succession of) the satellites (**206A-206E**). In various embodiments, the server(s) **202A** can control return (hand-back) of one or more devices from one or more of the satellites (**206A-206E**) to the wireless terrestrial network **202**.

[0037] As described herein, various embodiments provide a traffic management solution for heterogeneous terrestrial and satellite networks (e.g., 5G, 6G, subsequent generation) based on estimated UE transmit power. In various examples, an algorithm for making traffic management decisions can reside at the network, at the RAN, and/or behind the eNB/gNB (e.g., SON, RIC).

[0038] As described herein, various embodiments can collect UE-information of the UEs that are attached to the network. In various examples, the UE-information can comprise UE capability, UE device type, and/or UE frequency band capabilities. In addition, the decision-making algorithm can collect UE traffic demand, which can be obtained from the UE and/or from a network element (the traffic demand can comprise traffic type and/or bandwidth demand).

[0039] As described herein, various embodiments can provide an algorithm that uses the collected information to rank the UEs that are more suitable to be offloaded from the terrestrial network to the non-terrestrial network. This ranking can be based, for example, on each estimated UE transmit power and each current UE battery level. In one embodiment, a ranking can be as follows: [0040] Wearable UE or UE with low battery power engaged in heavy upload traffic=Low rank value (not suitable UE to offload to non-terrestrial network) [0041] Handheld UE with high battery power engaged into medium upload traffic=Middle rank value [0042] UE with high/constant battery power (e.g., DC Power) engaged into light upload traffic=High rank value (suitable UE to offload to non-terrestrial network)

[0043] As described herein, various embodiments can provide an algorithm that (after making one or more decisions) then mandates the terrestrial RAN to instruct one or more UEs to perform hand-over (or cell-reselection) to satellite cell(s), thus offloading traffic from a terrestrial network cell to a non-terrestrial network cell. In various examples, the offloading can be implemented by either adding higher priority to the Frequency Band or the Cell.ID that the satellite cells are using.

[0044] Referring now to FIG. 2B, various steps of a method **2000** according to an embodiment are shown. As seen in this FIG. 2B, step **2002** comprises obtaining first data associated with a satellite of a wireless non-terrestrial network, wherein the first data indicates a location of the satellite. Next, step **2004** comprises obtaining second data associated with a plurality of mobile devices that are respectively attached to a wireless terrestrial network, wherein the second data indicates a respective location of each of the plurality of mobile devices, a respective battery state of each of the plurality of mobile devices, and a respective transmit power that is required for each of the plurality of mobile devices to upload traffic to the wireless terrestrial network. Next, step **2006** comprises generating, based upon the location of the satellite indicated by the first data and upon the respective location of each of the plurality of mobile devices indicated by the second data, third data that indicates respective predicted transmit power that would be required for each of the plurality of mobile devices to upload traffic to the satellite. Next, step **2008** comprises selecting, based upon the respective battery state of each of the plurality of mobile devices, upon the respective transmit power that is required for each of the plurality of mobile devices to upload traffic to the wireless terrestrial network, and upon the respective predicted transmit power that would be required for each of the plurality of mobile devices to upload traffic to the satellite, at least one mobile device of the plurality of mobile devices to be instructed to upload traffic to the satellite, wherein the selecting results in at least one selected mobile device. Next, step **2010** comprises responsive to the selecting, sending a command to the at least one selected mobile device to upload the traffic to the satellite instead of to the wireless terrestrial network.

[0045] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2B, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0046] Referring now to FIG. 2C, various steps of a method **2100** according to an embodiment are shown. As seen in this FIG. 2C, step **2102** comprises obtaining first data associated with a first end-user device that is attached to a wireless terrestrial network, wherein the first data indicates a first location of the first end-user device and a first configuration of the first end-user device, and wherein the first location comprises a first current location, a first predicted future location, or any combination thereof. Next, step **2104** comprises obtaining second data associated with a second

end-user device that is attached to the wireless terrestrial network, wherein the second data indicates a second location of the second end-user device and a second configuration of the second end-user device, and wherein the second location comprises a second current location, a second predicted future location, or any combination thereof. Next, step **2106** comprises obtaining third data associated with a satellite of a wireless non-terrestrial network, wherein the third data indicates a third location of the satellite, and wherein the third location comprises a third current location, a third predicted future location, or any combination thereof. Next, step **2108** comprises generating, based upon the first location and upon the third location, fourth data that indicates a first predicted transmit power that would be required for the first end-user device to upload traffic to the satellite. Next, step **2110** comprises generating, based upon the second location and upon the third location, fifth data that indicates a second predicted transmit power that would be required for the second end-user device to upload traffic to the satellite. Next, step **2112** comprises selecting either the first end-user device or the second end-user device to be commanded to communicate with the satellite, wherein the selecting results in a selected end-user device, and wherein the selecting is based upon the first configuration of the first end-user device, upon the second configuration of the second end-user device, upon the first predicted transmit power that would be required for the first end-user device to upload traffic to the satellite, and upon the second predicted transmit power that would be required for the second end-user device to upload traffic to the satellite.

[0047] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2C, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0048] Referring now to FIG. 2D, various steps of a method **2200** according to an embodiment are shown. As seen in this FIG. 2D, step **2202** comprises obtaining, by a processing system comprising a processor, first information associated with a first end-user device that is attached to a wireless terrestrial network, wherein the first information indicates a first location of the first end-user device and a first battery state of the first end-user device, and wherein the first location comprises a first current location, a first predicted future location, or any combination thereof. Next, step **2204** comprises obtaining, by the processing system, second information associated with a second end-user device that is attached to the wireless terrestrial network, wherein the second information indicates a second location of the second end-user device and a second battery state of the second end-user device, and wherein the second location comprises a second current location, a second predicted future location, or any combination thereof. Next, step **2206** comprises obtaining, by the processing system, third information associated with a third end-user device that is attached to the wireless terrestrial network, wherein the third information indicates a third location of the second end-user device and a third battery state of the third end-user device, and wherein the third location comprises a third current location, a third predicted future location, or any combination thereof. Next, step **2208** comprises obtaining, by the processing system, fourth information indicative of a fourth location of a low earth orbit (LEO) satellite, wherein the LEO satellite is part of a constellation of LEO satellites forming a wireless non-terrestrial network, and wherein the fourth location comprises a fourth current location, a fourth predicted future location, or any combination thereof. Next, step **2210** comprises generating, by the processing system, based upon the first location and upon the fourth location, fifth information indicative of a first predicted power that would be required for the first end-user device to carry out first uplink transmissions to the LEO satellite. Next, step **2212** comprises generating, by the processing system, based upon the second location and upon the fourth location, sixth information indicative of a second predicted power that would be required for the second end-user device to carry out second uplink transmissions to the LEO satellite. Next, step **2214** comprises generating, by the processing system, based upon the third location and upon the fourth location, seventh information indicative of a third predicted

power that would be required for the third end-user device to carry out third uplink transmissions to the LEO satellite. Next, step **2216** comprises rank ordering, by the processing system, the first end-user device, the second end-user device and the third end-user device, wherein the rank ordering is based upon the first battery state, the first predicted power, the second battery state, the second predicted power, the third battery state, and the third predicted power, and wherein the rank ordering orders each of the first end-user device, the second end-user device, and the third end-user device with respect to being instructed to hand-over to the LEO satellite.

[0049] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2D, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0050] As described herein, various embodiments can provide systems and methods for traffic management based on uplink power for a non-terrestrial network (e.g., a 5G (or subsequent generation) network).

[0051] As described herein, various embodiments can provide an algorithm which ranks the UEs that are more suitable to be offloaded from terrestrial network to a non-terrestrial network. This ranking can be based on the estimated UE transmit power and current UE battery level. In general, the algorithm can avoid offloading wearable UEs (and/or UEs with low battery power) from the terrestrial network to the non-terrestrial network (such a non-terrestrial network is more likely to drain battery faster).

[0052] As described herein, various embodiments can be utilized by a wireless operator that is using (or is planning on using) non-terrestrial cells to provide extra coverage and/or capacity to a terrestrial wireless network.

[0053] As described herein, various embodiments can operate in the context of a terrestrial network that is aware of the availability of satellite coverage and/or one or more UEs that is aware of the satellite coverage.

[0054] As described herein, various embodiments can operate in the context of applying intelligence (e.g., artificial intelligence) to the question of which UE(s) to hand-over to a satellite from a terrestrial network and which UE(s) to keep with the terrestrial network.

[0055] As described herein, various embodiments can operate in the context of UEs that are communicating video data, audio data, video calls, audio calls, and/or streaming.

[0056] As described herein, various embodiments can operate in the context of power control/management for uplink communications, such as connecting from a cell phone to a base station (if the cell phone is close to the base station, the cell phone doesn't need to use too much power for transmission; on the other hand, if the cell phone is further from the base station then typically more transmit power by the cell phone would be needed).

[0057] As described herein, various embodiments can make hand-over decisions to a satellite based upon whether a given UE is engaged in heavy uploading, has been engaged in heavy uploading, is predicted to be engaged in heavy uploading or any combination thereof.

[0058] As described herein, various embodiments can provide an algorithm that collects information about the UEs, about the terrestrial network, about the satellite network, and then makes decisions based upon such data.

[0059] As described herein, various embodiments can provide an algorithm that minimizes battery drain for one or more UEs. The battery drain can be minimized for a number of UEs and can take into account current, past and/or predicted traffic patterns. In one example, the algorithm can essentially solve a maximize problem (wherein a constraint is preserving battery charge of one or more UEs). In one example, a plurality of UEs can be ranked or prioritized (e.g., prioritized to stay on the terrestrial network or prioritized to be sent to the satellite network).

[0060] As described herein, various embodiments can provide an algorithm that receives battery

charge (or battery level) data that is reported by each UE. In various examples, the UEs can report their respective battery charges/levels in real-time, or in quasi real-time.

[0061] As described herein, various embodiments can provide for a terrestrial network (e.g., a serving cell) to mandate (or force) one or more UEs to hand-over to a satellite network. In one embodiment, the terrestrial network can notify a user to perform (and/or allow) the hand-over.

[0062] As described herein, various embodiments can provide systems and methods for traffic management based on uplink power for non-terrestrial networks (e.g. 5G, 6G, subsequent generation).

[0063] As described herein, a cellular service provider could use certain conventional traffic management techniques to offload traffic from a terrestrial cellular network to a non-terrestrial network. Such conventional traffic management techniques were typically based on loading conditions and traditionally did not take into consideration the required UE transmit power (e.g., P_{sub}.PUSCH) and actual UE battery state. Therefore, it was possible that a UE offloaded to a non-terrestrial network could drain battery power more quickly as compared to a UE attached to a terrestrial network (e.g., if the UE is engaged into heavy uplink demand). Various embodiments provide traffic management techniques to address this issue.

[0064] As described herein, various embodiments can provide traffic management techniques applicable when offloading traffic from a terrestrial network to a non-terrestrial network based on UE transmit power.

[0065] Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, some or all of the subsystems and functions of system **200**, and/or some or all of the functions of methods **2000**, **2100**, **2200**. For example, virtualized communication network **300** can facilitate in whole or in part traffic management between a terrestrial network and a non-terrestrial network based upon user equipment battery state and uplink power requirements.

[0066] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0067] In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general-purpose processors or general-purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0068] As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and

associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it is elastic: so, the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0069] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized and might require special DSP code and analog front ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0070] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements do not typically need to forward large amounts of traffic, their workload can be distributed across a number of servers-each of which adds a portion of the capability, and which creates an elastic function with higher availability overall than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0071] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud or might simply orchestrate workloads supported entirely in NFV infrastructure from these third-party locations.

[0072] Turning now to FIG. **4**, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part traffic management between a terrestrial network and a non-terrestrial network based upon user equipment battery state and uplink power requirements.

[0073] Generally, program modules comprise routines, programs, components, data structures, etc.,

that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0074] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0075] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0076] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0077] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0078] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0079] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0080] With reference again to FIG. 4, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as

the processing unit **404**.

[0081] The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

[0082] The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high-capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0083] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0084] A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0085] A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0086] A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0087] The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote

computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0088] When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

[0089] When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0090] The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0091] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0092] Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part traffic management between a terrestrial network and a non-terrestrial network based upon user equipment battery state and uplink power requirements. In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks and can be considered carrier-side components as discussed

elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology (ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

[0093] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0094] In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0095] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550**

or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. **1(s)** that enhance wireless service coverage by providing more network coverage.

[0096] It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0097] In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0098] In order to provide a context for the various aspects of the disclosed subject matter, FIG. **5**, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0099] Turning now to FIG. **6**, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part traffic management between a terrestrial network and a non-terrestrial network based upon user equipment battery state and uplink power requirements.

[0100] The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, Wi-Fi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

[0101] The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as

monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0102] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0103] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

[0104] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0105] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

[0106] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, Wi-Fi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

[0107] Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

[0108] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise

clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0109] In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0110] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0111] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0112] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically managing traffic between a terrestrial network and a non-terrestrial network based upon user equipment battery state and uplink power requirements) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each user equipment device, each user, and/or each satellite. A classifier is a function that maps an input attribute vector, $x=(x_{\text{sub.1}}, x_{\text{sub.2}}, x_{\text{sub.3}}, x_{\text{sub.4}} \dots x_{\text{sub.n}})$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be

employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0113] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the user equipment device(s), user(s), and/or satellite(s) is to receive priority.

[0114] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0115] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various

embodiments.

[0116] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0117] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0118] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0119] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0120] As used herein, terms such as “data storage,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0121] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible.

Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the

claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0122] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0123] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0124] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A device, comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: obtaining first data associated with a satellite of a wireless non-terrestrial network, wherein the first data indicates a location of the satellite; obtaining second data associated with a plurality of mobile devices that are respectively attached to a wireless terrestrial network, wherein the second data indicates a respective location of each of the plurality of mobile devices, a respective battery state of each of the plurality of mobile devices, and a respective transmit power that is required for each of the plurality of mobile devices to upload traffic to the wireless terrestrial network; generating, based upon the location of the satellite indicated by the first data and upon the respective location of each of the plurality of mobile devices indicated by the second data, third data that indicates respective predicted transmit power that would be required for each of the plurality of mobile devices to upload traffic to the satellite;

selecting, based upon the respective battery state of each of the plurality of mobile devices, upon the respective transmit power that is required for each of the plurality of mobile devices to upload traffic to the wireless terrestrial network, and upon the respective predicted transmit power that would be required for each of the plurality of mobile devices to upload traffic to the satellite, at least one mobile device of the plurality of mobile devices to be instructed to upload traffic to the satellite, wherein the selecting results in at least one selected mobile device; and responsive to the selecting, sending a command to the at least one selected mobile device to upload the traffic to the satellite instead of to the wireless terrestrial network.

2. The device of claim 1, wherein the battery state of each of the plurality of mobile devices comprises: a respective battery power level; a respective indication that a battery of the mobile device is being charged by an external power source; or any combination thereof.

3. The device of claim 1, wherein the location of each of the plurality of mobile devices comprises: a respective geographic location; a respective latitude/longitude pair; a respective indication of a given network node via which the mobile device is attached to the wireless terrestrial network; or any combination thereof.

4. The device of claim 1, wherein each of the plurality of mobile devices being attached to the wireless terrestrial network comprises each of the plurality of mobile devices being: in a respective idle mode; in a respective connected mode; or any combination thereof.

5. The device of claim 1, wherein the command to the at least one selected mobile device to upload traffic to the satellite instead of to the wireless terrestrial network comprises a command to: perform a hand-over to the satellite; perform a cell-reselection to the satellite; or any combination thereof.

6. The device of claim 1, wherein: the location of the satellite comprises a current location; the first data further indicates a predicted future location of the satellite; and the generating the third data is further based upon both the location of the satellite and the future predicted location of the satellite.

7. The device of claim 1, wherein: the respective location of each of the plurality of mobile devices comprises a respective current location; the second data further indicates for each of the plurality of mobile devices a predicted future location; and the generating is further based upon both the current location of each of the plurality of mobile devices and the future predicted location of each of the plurality of mobile devices.

8. The device of claim 1, wherein: the first data further indicates one or more first wireless frequencies that the satellite is capable of using for communications; the second data further indicates for each of the plurality of mobile devices one or more respective second wireless frequencies that the mobile device is capable of using for communications; and the selecting is further based upon the one or more first wireless frequencies and the one or more respective second wireless frequencies.

9. The device of claim 8, wherein the at least one selected mobile device is selected based upon overlap between the one or more first wireless frequencies and at least one of the one or more respective second wireless frequencies.

10. The device of claim 1, wherein the wireless terrestrial network comprises one or more base stations, and each of the plurality of mobile devices is attached to one of the one or more base stations.

11. The device of claim 1, wherein: the wireless terrestrial network comprises an LTE cellular network, a fifth generation (5G) cellular network, a sixth generation (6G) cellular network, a subsequent generation cellular network, or any combination thereof; the satellite comprises a low earth orbit (LEO) satellite; and the satellite comprises a satellite communications cell or an operable portion thereof.

12. The device of claim 1, wherein each of the plurality of mobile devices comprises a respective wearable device, a respective smartwatch; a respective smartphone, a respective cellphone, a respective laptop computer, a respective notebook computer, a respective tablet, or any respective

combination thereof.

13. A non-transitory machine-readable medium comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: obtaining first data associated with a first end-user device that is attached to a wireless terrestrial network, wherein the first data indicates a first location of the first end-user device and a first configuration of the first end-user device, and wherein the first location comprises a first current location, a first predicted future location, or any combination thereof; obtaining second data associated with a second end-user device that is attached to the wireless terrestrial network, wherein the second data indicates a second location of the second end-user device and a second configuration of the second end-user device, and wherein the second location comprises a second current location, a second predicted future location, or any combination thereof; obtaining third data associated with a satellite of a wireless non-terrestrial network, wherein the third data indicates a third location of the satellite, and wherein the third location comprises a third current location, a third predicted future location, or any combination thereof; generating, based upon the first location and upon the third location, fourth data that indicates a first predicted transmit power that would be required for the first end-user device to upload traffic to the satellite; generating, based upon the second location and upon the third location, fifth data that indicates a second predicted transmit power that would be required for the second end-user device to upload traffic to the satellite; and selecting either the first end-user device or the second end-user device to be commanded to communicate with the satellite, wherein the selecting results in a selected end-user device, and wherein the selecting is based upon the first configuration of the first end-user device, upon the second configuration of the second end-user device, upon the first predicted transmit power that would be required for the first end-user device to upload traffic to the satellite, and upon the second predicted transmit power that would be required for the second end-user device to upload traffic to the satellite.

14. The non-transitory machine-readable medium of claim 13, wherein each of the first configuration and the second configuration comprises a respective: type of device; battery charge level; battery capacity; connection to an external source of power; or any combination thereof.

15. The non-transitory machine-readable medium of claim 13, wherein each type of device comprises a respective wearable device, a respective smartwatch; a respective smartphone, a respective cellphone, a respective laptop computer, a respective notebook computer, a respective tablet, or any respective combination thereof.

16. The non-transitory machine-readable medium of claim 13, wherein the operations further comprise responsive to the selecting, sending an instruction to the selected end-user device to upload the traffic to the satellite instead of to the wireless terrestrial network.

17. The non-transitory machine-readable medium of claim 13, wherein the selecting either the first end-user device or the second end-user device to be commanded to communicate with the satellite is further based upon a first transmit power that is required for the first end-user device to upload traffic to the wireless terrestrial network and upon a second transmit power that is required for the second end-user device to upload traffic to the wireless terrestrial network.

18. A method, comprising: obtaining, by a processing system comprising a processor, first information associated with a first end-user device that is attached to a wireless terrestrial network, wherein the first information indicates a first location of the first end-user device and a first battery state of the first end-user device, and wherein the first location comprises a first current location, a first predicted future location, or any combination thereof; obtaining, by the processing system, second information associated with a second end-user device that is attached to the wireless terrestrial network, wherein the second information indicates a second location of the second end-user device and a second battery state of the second end-user device, and wherein the second location comprises a second current location, a second predicted future location, or any combination thereof; obtaining, by the processing system, third information associated with a third

end-user device that is attached to the wireless terrestrial network, wherein the third information indicates a third location of the second end-user device and a third battery state of the third end-user device, and wherein the third location comprises a third current location, a third predicted future location, or any combination thereof; obtaining, by the processing system, fourth information indicative of a fourth location of a low earth orbit (LEO) satellite, wherein the LEO satellite is part of a constellation of LEO satellites forming a wireless non-terrestrial network, and wherein the fourth location comprises a fourth current location, a fourth predicted future location, or any combination thereof; generating, by the processing system, based upon the first location and upon the fourth location, fifth information indicative of a first predicted power that would be required for the first end-user device to carry out first uplink transmissions to the LEO satellite; generating, by the processing system, based upon the second location and upon the fourth location, sixth information indicative of a second predicted power that would be required for the second end-user device to carry out second uplink transmissions to the LEO satellite; generating, by the processing system, based upon the third location and upon the fourth location, seventh information indicative of a third predicted power that would be required for the third end-user device to carry out third uplink transmissions to the LEO satellite; and rank ordering, by the processing system, the first end-user device, the second end-user device and the third end-user device, wherein the rank ordering is based upon the first battery state, the first predicted power, the second battery state, the second predicted power, the third battery state, and the third predicted power, and wherein the rank ordering orders each of the first end-user device, the second end-user device, and the third end-user device with respect to being instructed to hand-over to the LEO satellite.

19. The method of claim 18, further comprising instructing whichever one of the first end-user device, the second end-user device, or the third end-user device that is ranked with a highest hand-over priority to communicate with the LEO satellite.

20. The method of claim 19, further comprising, after performing the instructing to communicate with the LEO satellite, further instructing whichever one of the first end-user device, the second end-user device, or the third end-user device that is ranked with a next-highest hand-over priority to also hand-over to the LEO satellite.
